

Supporting Information

Geography and life history mediate Pacific salmon responses to ocean warming and competition at sea

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1 Supplementary methods and results

1.1 Salmon data

We compiled data for pink, chum and sockeye salmon from a large public database of Pacific salmon spawner and recruitment time series (Greenberg (in prep)). These populations spanned 46.37 to 66.46 degrees in latitude and from 1948 to 2024. We discarded observations prior to 1950 due to concerns about data quality, and did not include time series that did not span at least brood years 1984-2014 and with minimal missing observation (i.e., > 80% of all years) (Figures S1-S3). We further excluded time series with known substantial hatchery influence to restrict our analysis to wild salmon populations. The resulting dataset included 66 sockeye populations, 28 chum salmon populations, and 33 pink salmon populations, grouped into three (chum) or four (sockeye and pink) large marine ecosystems/regions (Table ??). The most populations in a single region were West Coast sockeye (n=26) and the fewest were in the Bering Sea for pink salmon (n=4).

Unlike for other systems and species, estimates of spawner abundance for Fraser River were “effective female spawners” instead of total spawners, which accounts for the at times substantial pre-spawn mortality (i.e., mortality after arrival on the spawning grounds, but before spawning) and variable egg retention in this system. Additionally, return abundance estimates for Fraser River sockeye salmon account for en route mortality (i.e., mortality after freshwater entry, but before arrival on the spawning grounds), which can be considerable in certain years. While there is likely pre-spawn and en route mortality in other systems and species it is only systematically accounted for in the Fraser. Because of this, for modeling sockeye dynamics only, we allowed regional intrinsic productivity to arise from a common normal distribution that was allowed to differ from populations in other regions. In addition, for Fraser sockeye any confounding of freshwater temperature effects on return migration on estimates of marine temperature effects (e.g., in instances where they are correlated) in our analyses are accounted for by having adult recruitment explicitly account for freshwater migration mortality.

Sockeye can rear in freshwater for a variable length of time before migrating to sea, and both sockeye and chum can mature and return to spawn at variable ages. Where possible we used brood year specific estimates of age at ocean entry and return; when this data was not available we used estimates from adjacent years, or the mean across all years if only a few years had brood year specific estimates. In some instances for chum, there were no age data available, and so we assumed a fixed four-year life cycle, following Malick et al. (2017)

who found that assuming a fixed four year life cycle for chum introduced minimal bias into reconstructions of their population dynamics.

1.2 Sensitivity Analyses

Our analyses made a number of simplifying assumptions about processes that are hypothesized to influence Pacific salmon dynamics. Our intent was to quantify the statistical evidence for a relatively narrow set of hypotheses with strong a-priori justification (i.e., the model/hypotheses in the main text), and to then conduct sensitivity analyses to understand the extent to which our findings are sensitive to assumptions made during the analysis.

1.2.1 Alternative climate indices

Our analyses focused on coastal SST during early marine life as an index of ecological conditions during early marine life for each salmon population. This index has been repeatedly shown to correlate with pink, chum, and sockeye productivity across our study area Connors et al. (2020). However, previous analyses have also shown that the NPGO, that indexes sea surface height and is associated with primary production (e.g., nutrients, salinity, and chlorophyll-a.), can be associated with salmon productivity (Malick et al. 2017). To quantify the degree to which our inference about SST and pink salmon effects were confounded by other ocean basin scale processes (e.g., like those indexed by the NPGO) we repeated our analysis with the addition of the NPGO index. Specifically, we derived indices of the NPGO by averaging monthly values over the December-March period immediately prior to ocean entry, as in Mueter, Ware, and Peterman (2002) and Malick et al. (2017) and then included it as an additional covariate in either the stationary or “era” models. As with the SST index for sockeye the NPGO index was weighted by the proportion of each brood year cohort that outmigrated in each year. There was no strong positive correlation between the NPGO index and the SST or competitor covariates when all years were considered (-0.25 and 0.04 respectively). In contrast, our SST indices were moderately strongly correlated with another commonly considered ocean basin scale index, the PDO (0.4 across all years; 0.77 in the most recent time period), and so we did not consider it further in our analyses.

Stationary models that included the additional covariate indicated that the NPGO had a relatively small but typically positive relationship with productivity (Figures S7-S9), with the exception of Gulf of Alaska pink and chum where there was no evidence of a relationship (Figure S8 and S8), and Bering Sea chum where there was evidence of an inverse relationship. Importantly, considering the NPGO covariate had little effect on estimates of the effects of the SST and salmon competitor covariates. In the era models, there were larger magnitudes of positive NPGO effects in the most recent time period for sockeye (Figure S7), chum from more northern latitudes (Figure S8), and a negative effect for West Coast pink salmon (Figure S9). Overall, the spatial and temporal patterns we describe in the main text were insensitive to the simultaneous consideration of this additional covariate.

1.2.2 Alternative salmon competitor indices

To evaluate the sensitivity of our inference to alternative indices of inter- and intra-specific competition we considered four alternative competitor indices, in addition to the index of total North Pacific wide pink, chum, and sockeye salmon abundance we considered in the main text. These alternatives included: (1) the combined biomass of all sockeye, chum, and pink salmon in the North Pacific (based on conversion of adult abundance to biomass used in Ruggerone and Irvine (2018)), and the North Pacific wide abundance of just (2) pink, (3) chum, or (3) sockeye. Each represents a slightly different hypothesis about the nature of competition among salmon at sea, for example, whether competition occurs among all pink, chum, and sockeye salmon, or primarily with pink salmon, or whether biomass better indexes potential competitors than abundance. In addition, analyses that only considered an index of competitors that was the same species as the focal populations were effectively just examining evidence for intra-specific competition.

We quantified the sensitivity of our results to these alternative assumptions by substituting each alternative index for the total abundance of pink, chum and sockeye we considered as a competitor index into the stationary models described in eqn. 1-4 in the main text. The indices were calculated in the same way as the

original competitor index. We found that in general, while there were subtle differences in the estimated magnitude of competitor effects (and their interaction with SST), overall when estimated effects were strong in our baseline analysis they remained of a similar magnitude and direction when considering biomass instead of abundance but were often somewhat weaker when considering just individual species competitor indices (Figures S10-S12).

1.2.3 Detrended salmon competitor indices

The pink, chum, and sockeye salmon abundance time series we considered as an index of competition among salmon at sea exhibits an increasing time trend, in addition to high frequency year-to-year variation (Figure 1b in main text; Figure S13a). As a result it is possible that other unaccounted for processes that exhibit strong positive trends over the same period (e.g., marine mammal populations, ocean heat content) may confound our interpretation of a pink salmon competition “effect”. While the weight of evidence for the effect of competition with especially pink salmon on other salmon species is much stronger (Ruggerone et al. 2023) than for these other potential ecosystem changes, as a sensitivity analysis we also conducted an analysis with the temporal signal in the salmon time series removed. To do this we fit a General Additive Model to the North Pacific pink, chum, and sockeye salmon time series (Figure S13b) and used the resulting residuals of the model fit as an alternative, time trend removed, competitor index (Figure S13b). We then refit the stationary model (equation 1 in Main Text) with this alternative index to each of the sockeye, chum, and pink salmon spawner-recruitment data set.

Consistent with previous analyses that have fit models of salmon productivity to detrended salmon competitor indices (Ruggerone and Connors (2015), Connors et al. (2020)) statistical evidence of an inverse relationship between sockeye productivity and pink, chum and sockeye salmon abundance remained, though as expected since long term low frequency variation in the competitor time series was removed, the magnitude of predicted percentage change in recruits-per-spawner was smaller (Figure S14).

2 Figures

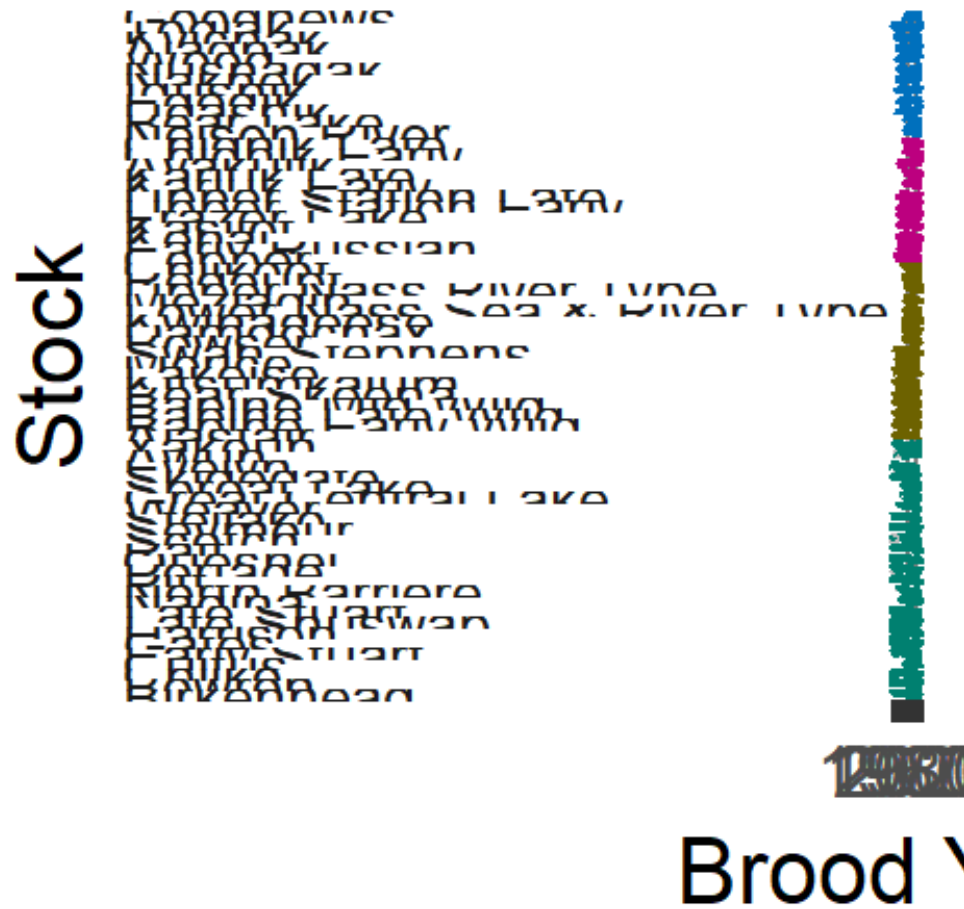


Figure S1: Sockeye salmon recruits-per-spawner timeseries used in the analysis. Missing brood years are in grey, regions are color coded (Table S1), and the temporal break point denoting the three “eras” considered are illustrated with vertical dashed lines.

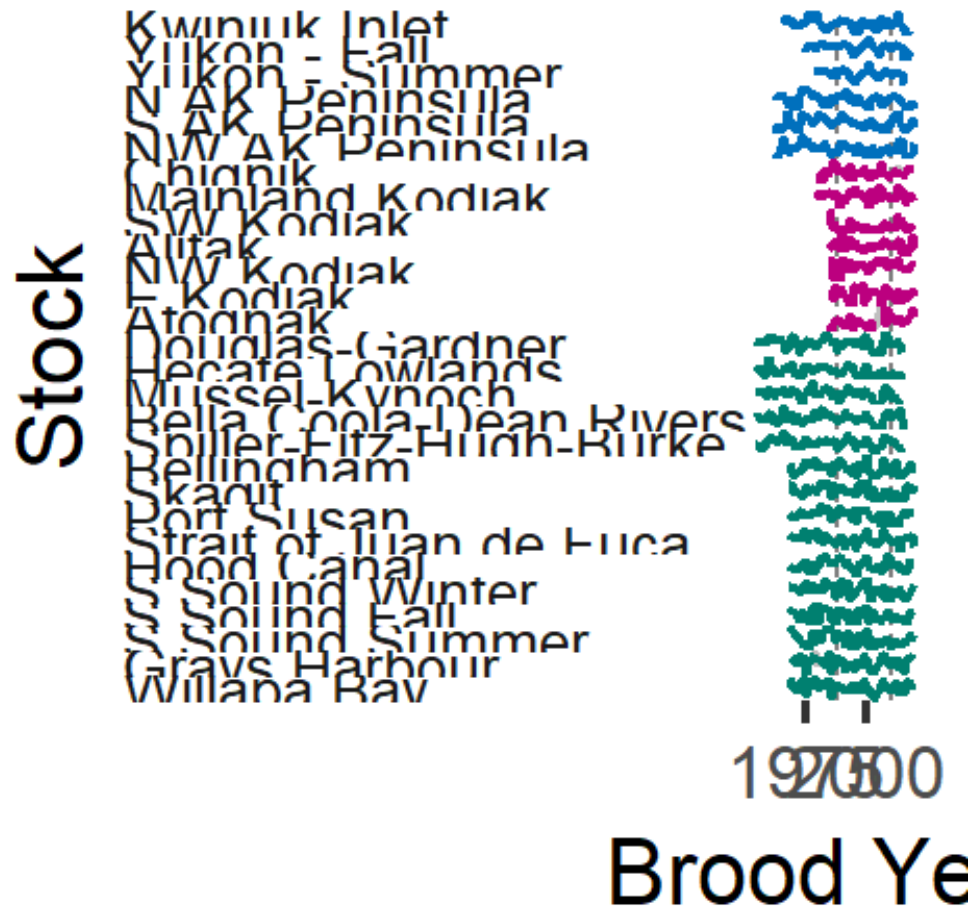


Figure S2: Chum salmon recruits-per-spawner timeseries used in the analysis. Missing brood years are in grey, regions are color coded (Table S1), and the temporal break point denoting the three “eras” considered are illustrated with vertical dashed lines.



Figure S3: Pink salmon recruits-per-spawner timeseries used in the analysis. Missing brood years are in grey, regions are color coded (Table S1), and the temporal break point denoting the three “eras” considered are illustrated with vertical dashed lines.

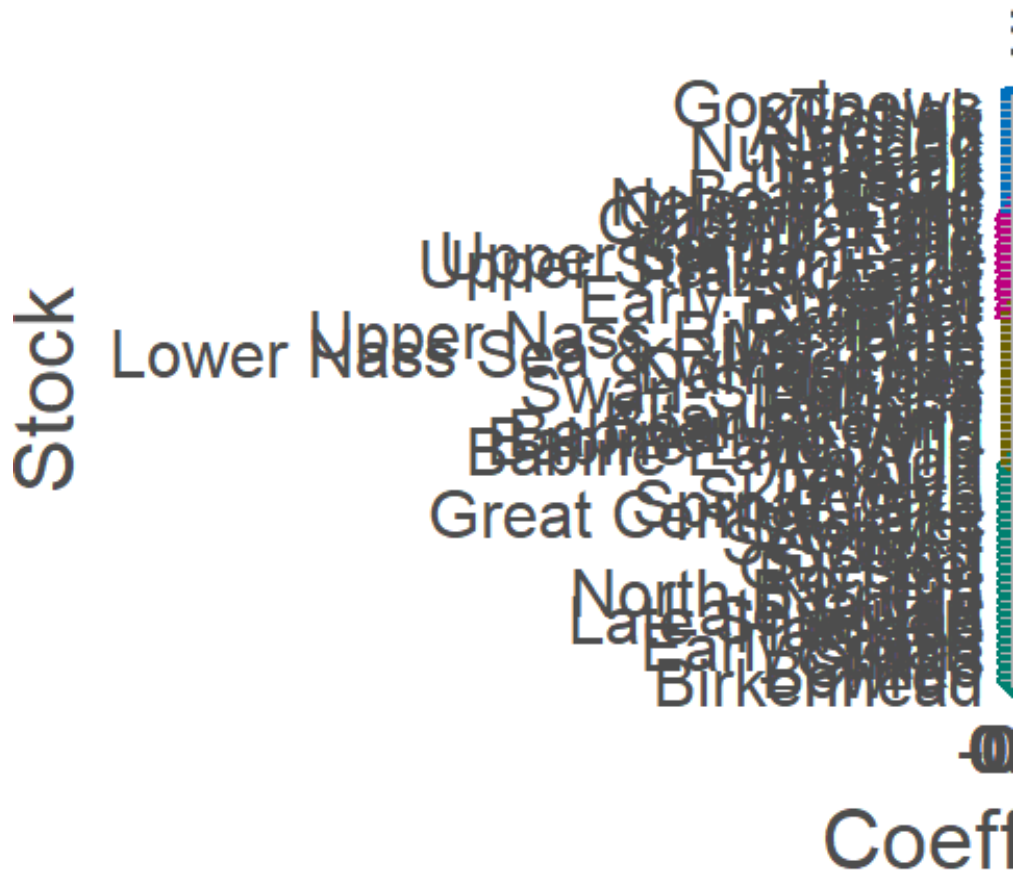


Figure S4: Sockeye posterior distribution summaries for SST and competition coefficients from the model described in the main text (equation 1). Filled points are posterior medians (points) and 95% credible intervals (horizontal lines) for the stock-specific effects of SST, and pink salmon competition, and open points are estimates from a simplified model that is fit individually and independently for each stock. Solid vertical lines give the posterior median for the ocean ecosystem level effects, which represent the mean effect across all stocks in an ocean ecosystem. Shaded box give the 95% credibility interval for the ecosystem-level effect with dashed lines indicating zero effect.

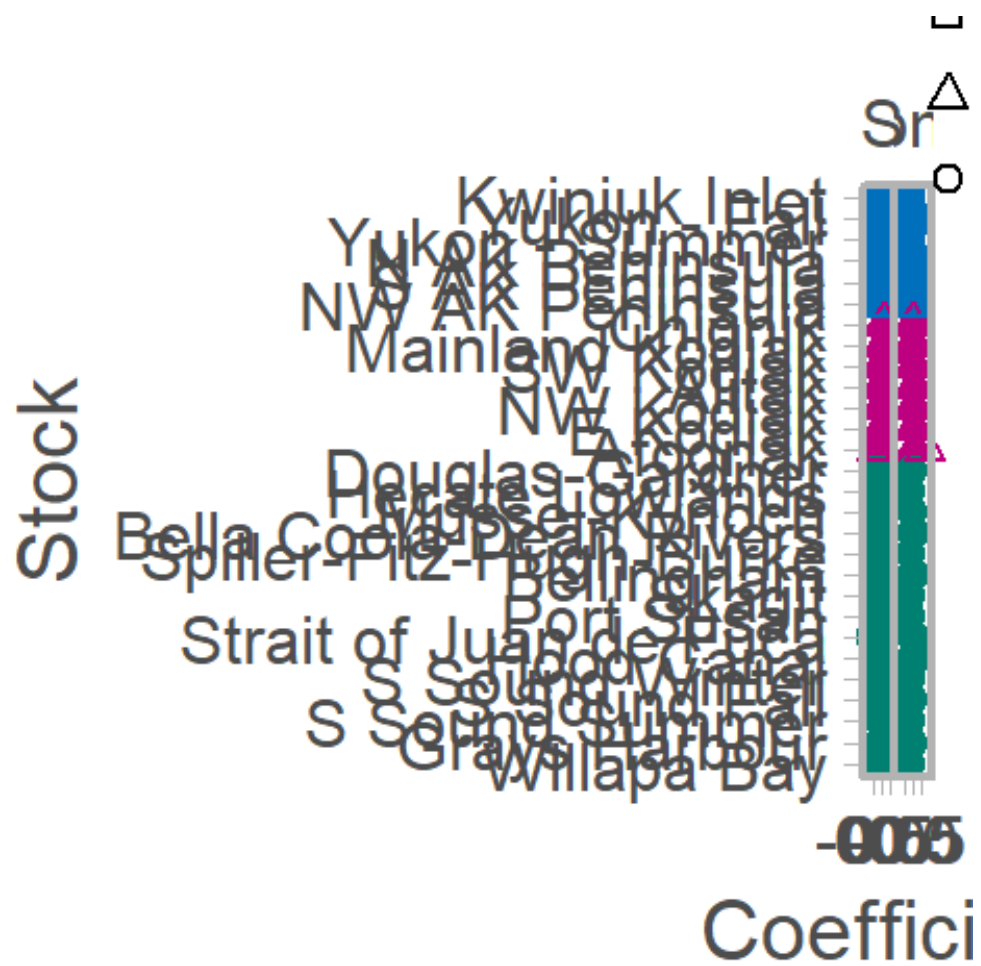


Figure S5: Same as Figure S4 but for chum salmon stocks.

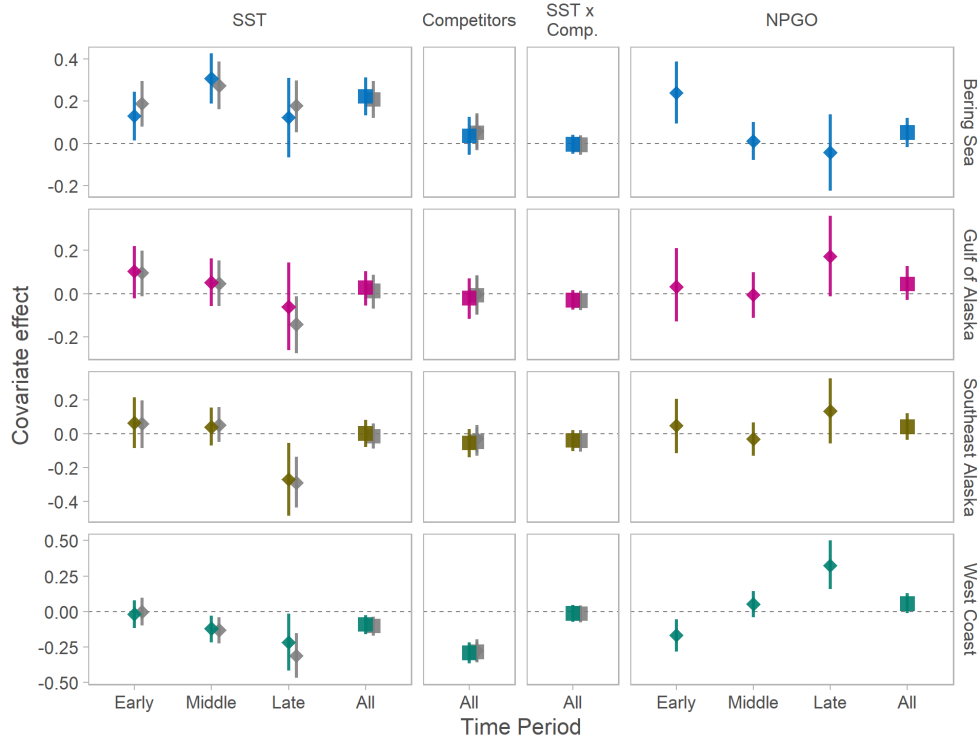


Figure S7: Posterior distribution summaries for covariate effects on sockeye salmon productivity when including the NPGO as an additional covariate in models (coloured points and lines) compared to original models with only SST, salmon competitors, and their interaction (grey points and lines). Points are medians of posterior distributions, lines are 95% credible intervals. Effect sizes estimated from time-varying models (pre-defined break points) are indicated with diamond-shaped points and labeled by time period: “Early” (1960-1988), “Middle” (1989-2010), or “Late” (2011-2016). Estimates from stationary models considering all years are indicated with square points and labeled “All”.

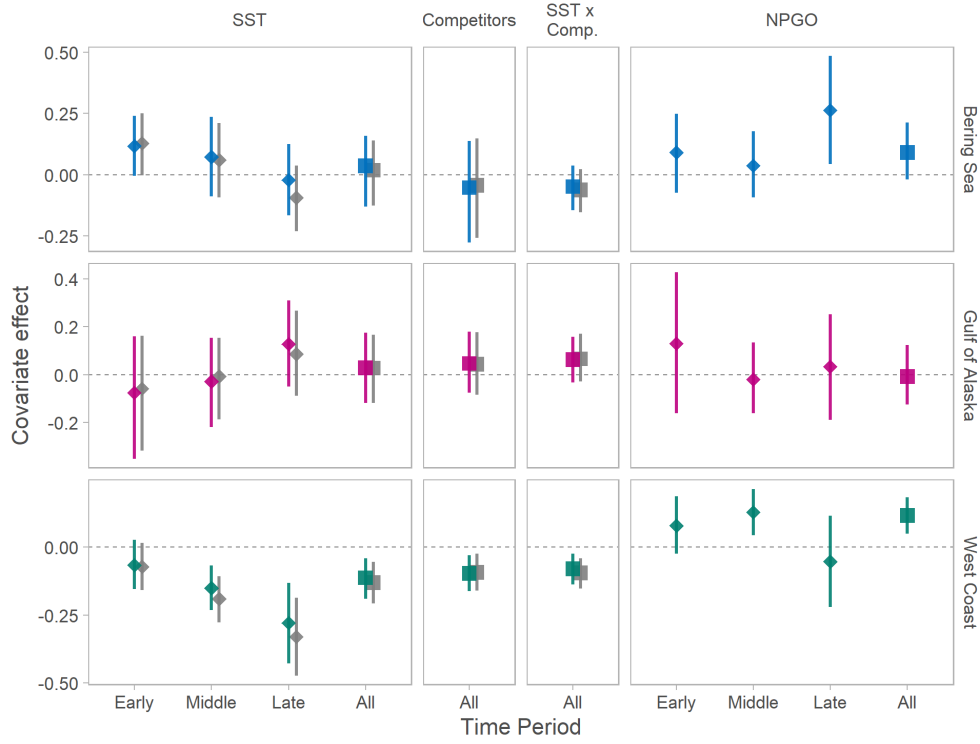


Figure S8: Same as Figure S7 but for chum salmon stocks.

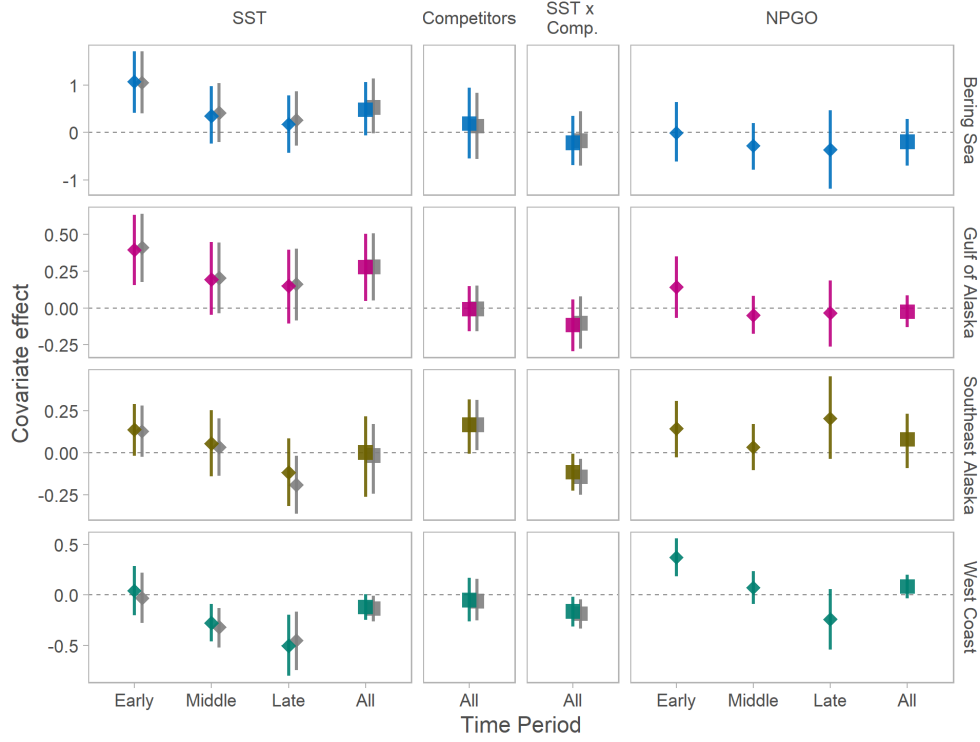


Figure S9: Same as Figure S7 but for pink salmon stocks.

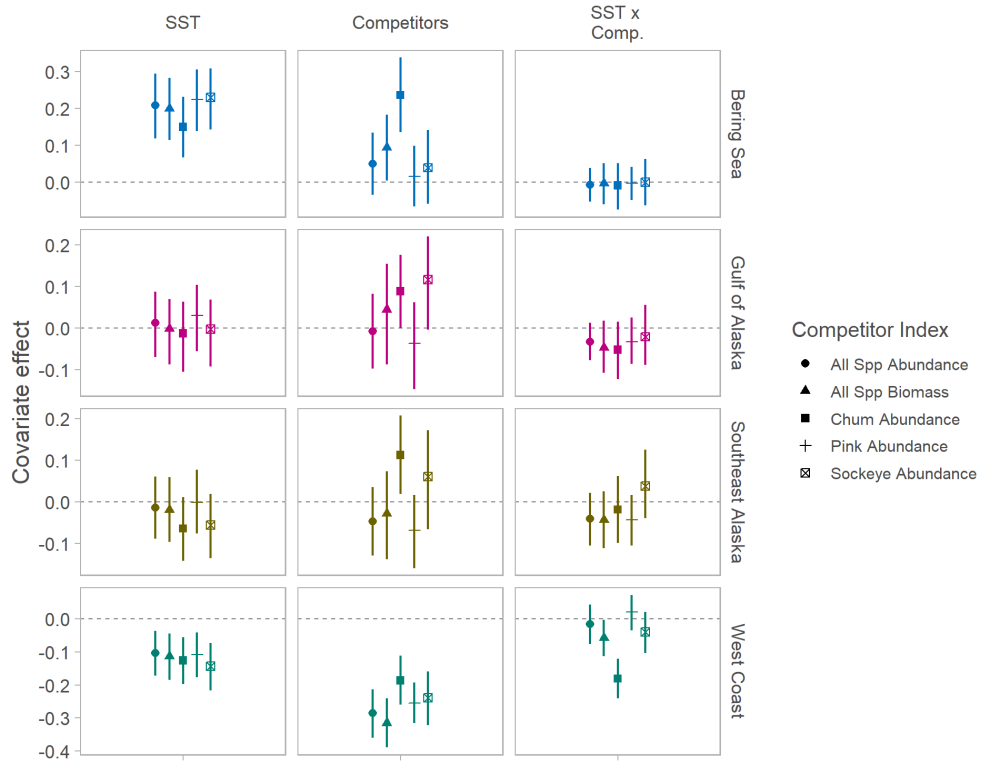


Figure S10: Posterior distribution summaries for covariate effects on sockeye salmon productivity across alternative indices of potential salmon competitors. Points are posterior means and vertical lines are 95% credible intervals.

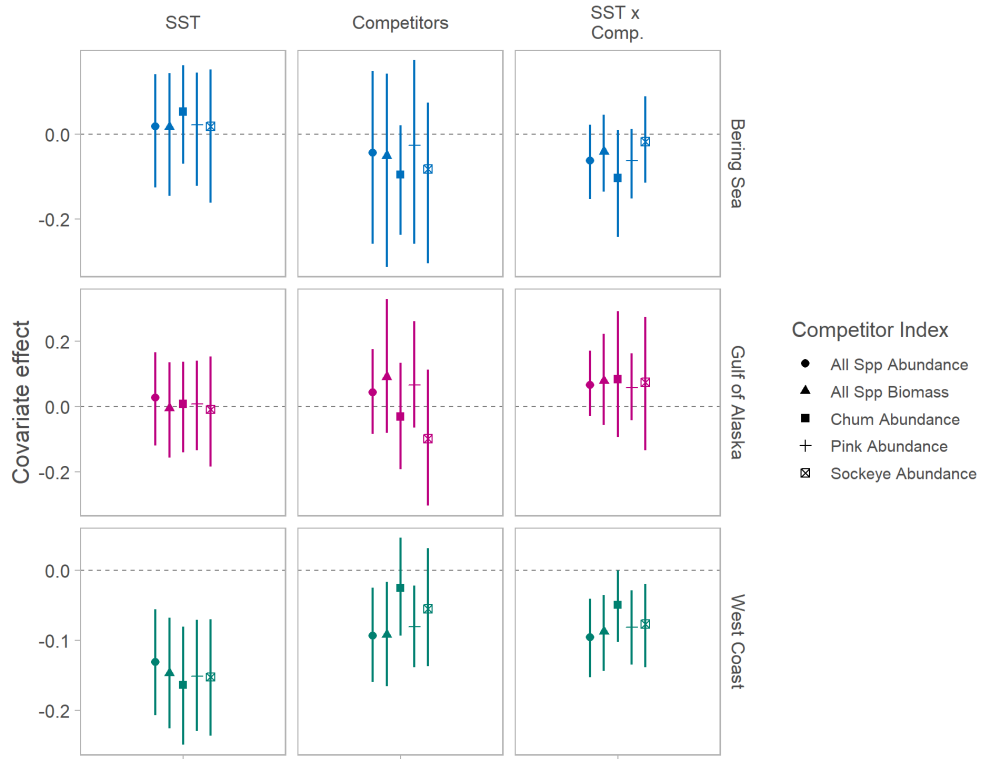


Figure S11: Posterior distribution summaries for covariate effects on chum salmon productivity across alternative indices of potential salmon competitors. Points are posterior means and vertical lines are 95% credible intervals.

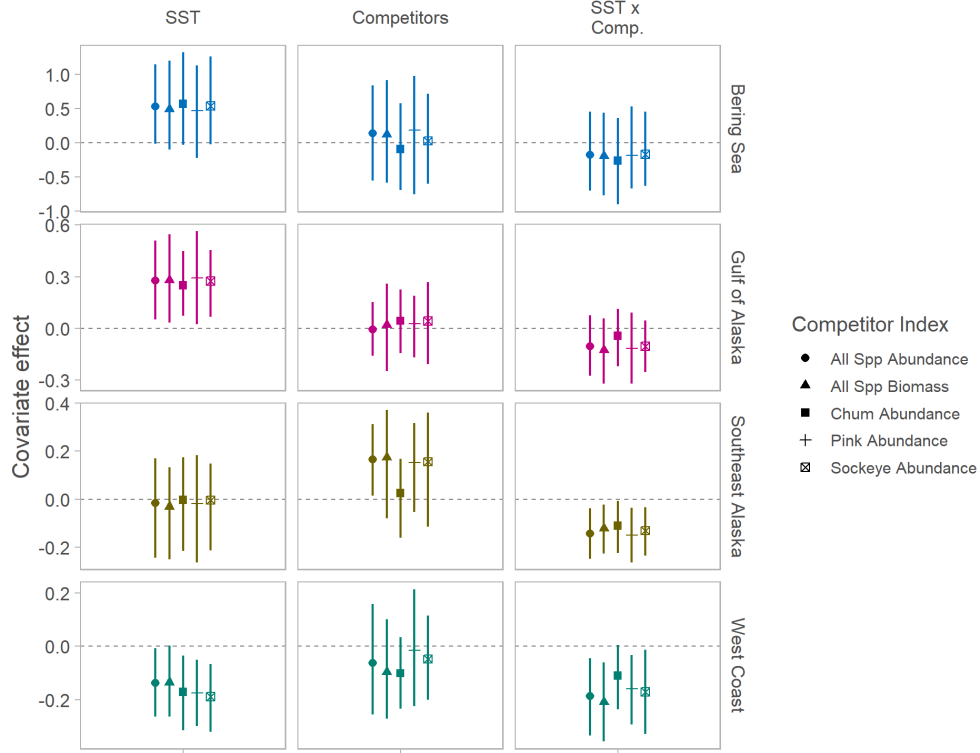


Figure S12: Posterior distribution summaries for covariate effects on pink salmon productivity across alternative indices of potential salmon competitors. Points are posterior means and vertical lines are 95% credible intervals.

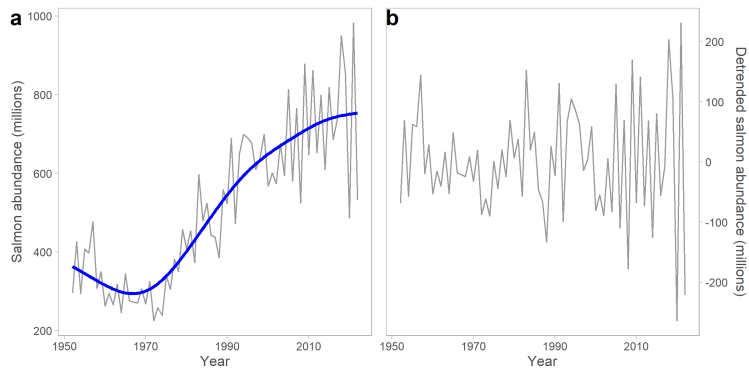


Figure S13: (a) North Pacific pink, chum, and sockeye salmon abundance over time (light grey) and GAM model fit to time series (blue) with resulting detrended competitor index (i.e., residuals of GAM model fit) show in (b).

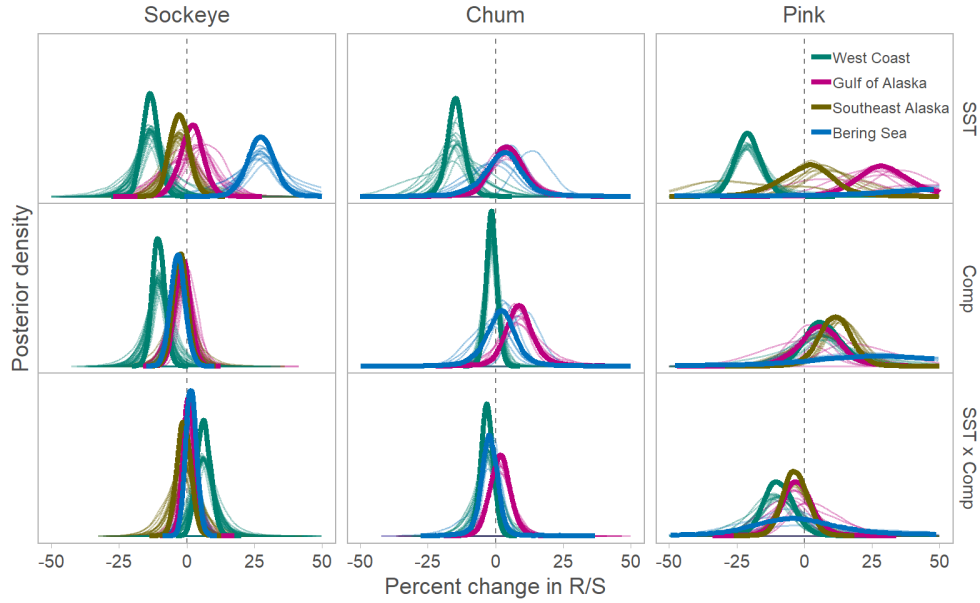


Figure S14: Effect of detrending salmon competitor index on estimates of impact of ocean warming and competition at sea. Posterior probability distributions of the predicted effect of SST and detrended competitors (Figure S13b) on sockeye, chum and pink salmon productivity (percentage change in adult recruits per spawner) by ocean ecosystem (same as Figure 2 in Main Text).

3 Tables

Table S1. Populations included in analysis, with region assigned in analysis, years of data covered, and coordinates of ocean entry point.

Species
Stock
Ocean Region
First BY
Last BY
Lat.
Lon.
Sockeye
Birkenhead
West Coast
1950
2015
49.12000
-123.0600
Sockeye
Bowron

West Coast

1950

2015

49.12000

-123.0600

Sockeye

Chilko

West Coast

1950

2015

49.12000

-123.0600

Sockeye

Cultus

West Coast

1950

2015

49.12000

-123.0600

Sockeye

Early Stuart

West Coast

1950

2015

49.12000

-123.0600

Sockeye

Gates

West Coast

1952

2015

49.12000

-123.0600

Sockeye

Harrison

West Coast

1950
2015
49.12000
-123.0600
Sockeye
Late Shuswap
West Coast
1950
2015
49.12000
-123.0600
Sockeye
Late Stuart
West Coast
1950
2015
49.12000
-123.0600
Sockeye
Nadina
West Coast
1973
2015
49.12000
-123.0600
Sockeye
North Barriere
West Coast
1958
2015
49.12000
-123.0600
Sockeye
Pitt
West Coast
1950

2015
49.12000
-123.0600
Sockeye
Portage
West Coast
1951
2015
49.12000
-123.0600
Sockeye
Quesnel
West Coast
1950
2015
49.12000
-123.0600
Sockeye
Raft
West Coast
1950
2015
49.12000
-123.0600
Sockeye
Scotch
West Coast
1952
2015
49.12000
-123.0600
Sockeye
Seymour
West Coast
1950
2015

49.12000
-123.0600
Sockeye
Stellako
West Coast
1950
2015
49.12000
-123.0600
Sockeye
Weaver
West Coast
1966
2015
49.12000
-123.0600
Sockeye
Great Central Lake
West Coast
1977
2015
49.24000
-124.8000
Sockeye
Sproat Lake
West Coast
1977
2015
49.24000
-124.8000
Sockeye
Canoona
West Coast
1975
2014
53.08000

-128.5550
Sockeye
Skidegate
West Coast
1956
2015
53.16700
-131.7890
Sockeye
Evelyn
West Coast
1954
2014
53.56000
-128.9580
Sockeye
Awun
West Coast
1957
2015
53.66500
-132.5230
Sockeye
Yakoun
West Coast
1957
2015
53.68200
-132.2350
Sockeye
Alastair
Southeast Alaska
1960
2015
54.22000
-129.8310

Sockeye
Babine Early Wild
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Babine Late Wild
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Babine Mid Wild
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Bear-Skeena
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Kitsumkalum
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye

Lakelse
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Morice
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Swan-Stephens
Southeast Alaska
1960
2015
54.22000
-129.8310
Sockeye
Bowser
Southeast Alaska
1982
2015
54.99000
-130.0200
Sockeye
Damdochax
Southeast Alaska
1982
2015
54.99000
-130.0200
Sockeye
Kwinageese

Southeast Alaska
1982
2015
54.99000
-130.0200
Sockeye
Lower Nass Sea & River Type
Southeast Alaska
1982
2015
54.99000
-130.0200
Sockeye
Meziadin
Southeast Alaska
1982
2015
54.99000
-130.0200
Sockeye
Upper Nass River Type
Southeast Alaska
1982
2015
54.99000
-130.0200
Sockeye
Redoubt
Southeast Alaska
1982
2015
56.90000
-135.3300
Sockeye
Chilkoot
Southeast Alaska

1976
2015
58.92000
-135.2300
Sockeye
Copper
Gulf of Alaska
1961
2015
60.46000
-144.9600
Sockeye
Early Russian
Gulf of Alaska
1965
2015
60.54000
-151.2800
Sockeye
Kenai
Gulf of Alaska
1968
2015
60.54000
-151.2800
Sockeye
Kasilof
Gulf of Alaska
1968
2015
60.39000
-151.3000
Sockeye
Upper Station Early
Gulf of Alaska
1969

2015
57.06000
-154.3600
Sockeye
Upper Station Late
Gulf of Alaska
1970
2015
57.06000
-154.3600
Sockeye
Karluk Early
Gulf of Alaska
1982
2015
57.57000
-154.4500
Sockeye
Karluk Late
Gulf of Alaska
1982
2015
57.57000
-154.4500
Sockeye
Ayakulik
Gulf of Alaska
1966
2015
57.20000
-154.5400
Sockeye
Chignik Early
Gulf of Alaska
1983
2015

56.28000
-158.6400
Sockeye
Chignik Late
Gulf of Alaska
1983
2015
56.28000
-158.6400
Sockeye
Nelson
Bering Sea
1982
2015
55.96000
-161.2800
Sockeye
Bear
Bering Sea
1982
2015
56.16000
-160.4500
Sockeye
Kvichak
Bering Sea
1963
2014
57.50000
-157.6000
Sockeye
Alagnak
Bering Sea
1963
2014
58.25000

-157.3800
Sockeye
Wood
Bering Sea
1963
2014
58.68000
-158.8000
Sockeye
Naknek
Bering Sea
1963
2014
58.78000
-157.0800
Sockeye
Igushik
Bering Sea
1963
2014
58.82000
-158.6200
Sockeye
Nushagak
Bering Sea
1963
2014
58.82000
-158.4900
Sockeye
Egegik
Bering Sea
1963
2014
58.85000
-157.0300

Sockeye
Ugashik
Bering Sea
1963
2014
58.85000
-158.5000
Sockeye
Togiak
Bering Sea
1956
2015
58.95000
-160.4500
Sockeye
Goodnews
Bering Sea
1981
2015
59.11000
-161.6200
Chum
Willapa Bay
West Coast
1968
2016
46.37330
-123.9575
Chum
Grays Harbour
West Coast
1969
2016
46.95083
-124.0497
Chum

S Sound Summer

West Coast

1968

2016

47.04200

-122.8350

Chum

S Sound Fall

West Coast

1968

2016

47.04200

-122.8350

Chum

S Sound Winter

West Coast

1968

2016

47.04200

-122.8350

Chum

Hood Canal

West Coast

1968

2016

47.56440

-122.9956

Chum

Strait of Juan de Fuca

West Coast

1968

2016

48.16200

-123.5900

Chum

Port Susan

West Coast
1968
2016
48.23861
-122.3758
Chum
Skagit
West Coast
1968
2016
48.38750
-122.3656
Chum
Bellingham
West Coast
1968
2016
48.77500
-122.5978
Chum
Spiller-Fitz-Hugh-Burke
West Coast
1954
2016
52.16000
-128.1400
Chum
Bella Coola-Dean Rivers
West Coast
1954
2016
52.38000
-126.7800
Chum
Mussel-Kynoch
West Coast

1954
2015
52.61000
-128.5200
Chum
Hecate Lowlands
West Coast
1954
2016
53.42000
-129.2500
Chum
Douglas-Gardner
West Coast
1954
2015
53.96000
-128.6800
Chum
Afognak
Gulf of Alaska
1984
2019
58.40000
-152.0000
Chum
E Kodiak
Gulf of Alaska
1984
2019
57.08986
-153.1879
Chum
NW Kodiak
Gulf of Alaska
1984

2019
57.85684
-153.8829
Chum
Alitak
Gulf of Alaska
1984
2019
56.88528
-154.1217
Chum
SW Kodiak
Gulf of Alaska
1984
2019
57.26451
-154.6132
Chum
Mainland Kodiak
Gulf of Alaska
1979
2019
58.03333
-154.6644
Chum
Chignik
Gulf of Alaska
1980
2019
56.59639
-157.9933
Chum
NW AK Peninsula
Bering Sea
1962
2019

54.98200
-163.3700
Chum
S AK Peninsula
Bering Sea
1962
2019
55.50000
-161.0000
Chum
N AK Peninsula
Bering Sea
1962
2019
56.46000
-159.6800
Chum
Yukon - Summer
Bering Sea
1978
2017
62.55000
-165.0000
Chum
Yukon - Fall
Bering Sea
1974
2019
62.55000
-165.0000
Chum
Kwiniuk Inlet
Bering Sea
1965
2018
64.72000

-161.9600
Pink
Nisqually-Odd
West Coast
1959
2019
47.10100
-122.7060
Pink
S South Misc.-Odd
West Coast
1959
2019
47.16000
-122.9100
Pink
Puyallup-Odd
West Coast
1959
2019
47.26972
-122.4272
Pink
Hood Canal-Odd
West Coast
1959
2019
47.56440
-122.9956
Pink
Snohomish-Odd
West Coast
1959
2019
48.02139
-122.2072

Pink
Dungeness-Odd
West Coast
1959
2019
48.15080
-123.1333
Pink
Stillaguamish-Odd
West Coast
1959
2019
48.23861
-122.3758
Pink
Skagit-Odd
West Coast
1959
2019
48.38750
-122.3656
Pink
Nooksack-Odd
West Coast
1959
2019
48.77500
-122.5978
Pink
Fraser-Odd
West Coast
1961
2019
49.12333
-123.0569
Pink

Hecate Lowlands-Even
 West Coast
 1954
 2018
 52.45000
 -129.1000
 Pink
 Hecate Strait-Fjords-Even
 West Coast
 1954
 2018
 52.45000
 -129.1000
 Pink
 N Haida Gwaii-Even
 Southeast Alaska
 1954
 2018
 54.09000
 -132.1400
 Pink
 S Southeast Alaska-Even
 Southeast Alaska
 1960
 2020
 55.71556
 -131.8714
 Pink
 S Southeast Alaska-Odd
 Southeast Alaska
 1961
 2019
 55.71556
 -131.8714
 Pink
 N Southeast Alaska Outside-Even

Southeast Alaska
 1960
 2020
 57.17694
 -135.7869
 Pink
 N Southeast Alaska Outside-Odd
 Southeast Alaska
 1961
 2019
 57.17694
 -135.7869
 Pink
 N Southeast Alaska Inside-Even
 Southeast Alaska
 1960
 2020
 57.57528
 -134.7503
 Pink
 N Southeast Alaska Inside-Odd
 Southeast Alaska
 1961
 2019
 57.57528
 -134.7503
 Pink
 Auke Creek-Even
 Southeast Alaska
 1978
 2016
 58.37600
 -134.6500
 Pink
 Auke Creek-Odd
 Southeast Alaska

1979
2015
58.37600
-134.6500
Pink
Prince William Sound-Even
Gulf of Alaska
1960
2016
60.50000
-147.0000
Pink
Prince William Sound-Odd
Gulf of Alaska
1961
2017
60.50000
-147.0000
Pink
Kodiak-Even
Gulf of Alaska
1978
2020
58.03333
-154.6644
Pink
Kodiak-Odd
Gulf of Alaska
1979
2019
58.03333
-154.6644
Pink
Chignik-Even
Gulf of Alaska
1980

2020
56.38528
-158.2261
Pink
Chignik-Odd
Gulf of Alaska
1981
2019
56.38528
-158.2261
Pink
S AKPen-Even
Gulf of Alaska
1962
2020
55.00000
-161.0000
Pink
S AKPen-Odd
Gulf of Alaska
1963
2019
55.00000
-161.0000
Pink
NW AK Peninsula-Even
Bering Sea
1962
2020
54.98200
-163.3700
Pink
NW AK Peninsula-Odd
Bering Sea
1963
2019

54.98200
 -163.3700
 Pink
 Kwiniuk-Even
 Bering Sea
 1966
 2020
 64.68000
 -162.0100
 Pink
 Kwiniuk-Odd
 Bering Sea
 1965
 2019
 64.68000
 -162.0100

Table S2. Estimated mean effect sizes by region for SST and Competitor covariates, and interaction term, from stationary models.

Species
Ocean Region
Coefficient
Lower CI
Mean
Upper CI
Mean % change in R/S
sockeye
West Coast
SST
-0.17
-0.10
-0.04
-9.76
sockeye
Southeast Alaska
SST
-0.09

-0.01
0.06
-1.43
sockeye
Gulf of Alaska
SST
-0.07
0.01
0.09
1.25
sockeye
Bering Sea
SST
0.12
0.21
0.29
23.21
sockeye
West Coast
Comp
-0.36
-0.29
-0.21
-24.82
sockeye
Southeast Alaska
Comp
-0.13
-0.05
0.03
-4.53
sockeye
Gulf of Alaska
Comp
-0.10
-0.01

0.08
-0.80
sockeye
Bering Sea
Comp
-0.03
0.05
0.13
5.10
sockeye
West Coast
SST x Comp
-0.08
-0.01
0.04
-1.46
sockeye
Southeast Alaska
SST x Comp
-0.10
-0.04
0.02
-4.03
sockeye
Gulf of Alaska
SST x Comp
-0.08
-0.03
0.01
-3.24
sockeye
Bering Sea
SST x Comp
-0.05
-0.01
0.04

-0.65
chum
West Coast
SST
-0.21
-0.13
-0.06
-12.29
chum
Gulf of Alaska
SST
-0.12
0.03
0.17
2.73
chum
Bering Sea
SST
-0.13
0.02
0.14
1.91
chum
West Coast
Comp
-0.16
-0.09
-0.02
-8.91
chum
Gulf of Alaska
Comp
-0.08
0.04
0.18
4.48

chum
Bering Sea
Comp
-0.26
-0.04
0.15
-4.22
chum
West Coast
SST x Comp
-0.15
-0.10
-0.04
-9.10
chum
Gulf of Alaska
SST x Comp
-0.03
0.07
0.17
6.76
chum
Bering Sea
SST x Comp
-0.15
-0.06
0.02
-5.99
pink
West Coast
SST
-0.26
-0.14
-0.01
-12.76
pink

Southeast Alaska

SST

-0.24

-0.02

0.17

-1.64

pink

Gulf of Alaska

SST

0.05

0.28

0.51

32.11

pink

Bering Sea

SST

-0.02

0.52

1.14

69.04

pink

West Coast

Comp

-0.25

-0.06

0.16

-5.94

pink

Southeast Alaska

Comp

0.02

0.17

0.31

18.15

pink

Gulf of Alaska

Comp
-0.16
-0.01
0.15
-0.60
pink
Bering Sea
Comp
-0.56
0.13
0.84
14.35
pink
West Coast
SST x Comp
-0.33
-0.19
-0.04
-16.99
pink
Southeast Alaska
SST x Comp
-0.25
-0.14
-0.04
-13.30
pink
Gulf of Alaska
SST x Comp
-0.27
-0.10
0.08
-9.89
pink
Bering Sea
SST x Comp

-0.71

-0.18

0.45

-16.15

Table S3. Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models.

Species

Stock

Ocean Region

Coefficient

Lower CI

Mean

Upper CI

Mean % change in R/S

sockeye

Birkenhead

West Coast

SST

-0.26

-0.11

0.01

-10.85

sockeye

Bowron

West Coast

SST

-0.24

-0.10

0.04

-9.86

sockeye

Chilko

West Coast

SST

-0.24

-0.10

0.00
-9.94
sockeye
Cultus
West Coast
SST
-0.40
-0.15
-0.02
-14.03
sockeye
Early Stuart
West Coast
SST
-0.25
-0.12
-0.02
-11.33
sockeye
Gates
West Coast
SST
-0.29
-0.12
0.00
-10.90
sockeye
Harrison
West Coast
SST
-0.23
-0.08
0.08
-7.91
sockeye
Late Shuswap

West Coast
SST
-0.33
-0.14
-0.02
-12.73
sockeye
Late Stuart
West Coast
SST
-0.22
-0.09
0.07
-8.70
sockeye
Nadina
West Coast
SST
-0.20
-0.08
0.08
-7.81
sockeye
North Barriere
West Coast
SST
-0.27
-0.12
0.01
-11.65
sockeye
Pitt
West Coast
SST
-0.18
-0.08

0.07
-7.55
sockeye
Portage
West Coast
SST
-0.28
-0.12
0.01
-11.41
sockeye
Quesnel
West Coast
SST
-0.27
-0.12
0.00
-11.35
sockeye
Raft
West Coast
SST
-0.22
-0.10
0.03
-9.57
sockeye
Scotch
West Coast
SST
-0.30
-0.13
-0.01
-11.76
sockeye
Seymour

West Coast

SST

-0.26

-0.11

0.01

-10.82

sockeye

Stellako

West Coast

SST

-0.25

-0.12

-0.01

-11.32

sockeye

Weaver

West Coast

SST

-0.27

-0.11

0.04

-10.28

sockeye

Great Central Lake

West Coast

SST

-0.26

-0.09

0.05

-8.69

sockeye

Sproat Lake

West Coast

SST

-0.25

-0.10

0.03
-9.28
sockeye
Skidegate
West Coast
SST
-0.21
-0.09
0.02
-8.92
sockeye
Evelyn
West Coast
SST
-0.29
-0.12
0.01
-11.03
sockeye
Awun
West Coast
SST
-0.14
0.00
0.27
0.46
sockeye
Yakoun
West Coast
SST
-0.16
-0.06
0.11
-5.73
sockeye
Alastair

Southeast Alaska
 SST
 -0.13
 -0.02
 0.10
 -1.57
 sockeye
 Babine Early Wild
 Southeast Alaska
 SST
 -0.19
 -0.05
 0.06
 -4.59
 sockeye
 Babine Late Wild
 Southeast Alaska
 SST
 -0.21
 -0.05
 0.07
 -5.00
 sockeye
 Babine Mid Wild
 Southeast Alaska
 SST
 -0.20
 -0.05
 0.07
 -4.62
 sockeye
 Bear-Skeena
 Southeast Alaska
 SST
 -0.10
 0.01

0.13
0.91
sockeye
Kitsumkalum
Southeast Alaska
SST
-0.07
0.02
0.13
1.83
sockeye
Lakelse
Southeast Alaska
SST
-0.12
-0.01
0.09
-1.39
sockeye
Morice
Southeast Alaska
SST
-0.07
0.04
0.22
3.79
sockeye
Swan-Stephens
Southeast Alaska
SST
-0.12
-0.01
0.10
-1.24
sockeye
Bowser

Southeast Alaska

SST

-0.18

-0.04

0.06

-4.19

sockeye

Damdochax

Southeast Alaska

SST

-0.14

-0.03

0.07

-2.87

sockeye

Kwinageese

Southeast Alaska

SST

-0.14

-0.01

0.13

-1.14

sockeye

Lower Nass Sea & River Type

Southeast Alaska

SST

-0.10

0.02

0.18

1.61

sockeye

Meziadin

Southeast Alaska

SST

-0.14

-0.02

0.07
-2.27
sockeye
Upper Nass River Type
Southeast Alaska
SST
-0.14
-0.02
0.09
-2.03
sockeye
Redoubt
Southeast Alaska
SST
-0.15
-0.02
0.11
-2.06
sockeye
Chilkoot
Southeast Alaska
SST
-0.10
0.01
0.14
0.84
sockeye
Copper
Gulf of Alaska
SST
0.00
0.08
0.18
8.64
sockeye
Early Russian

Gulf of Alaska

SST

-0.08

0.04

0.15

3.60

sockeye

Kenai

Gulf of Alaska

SST

-0.17

-0.03

0.07

-3.44

sockeye

Kasilof

Gulf of Alaska

SST

-0.04

0.04

0.13

4.46

sockeye

Frazer Lake

Gulf of Alaska

SST

-0.20

-0.02

0.10

-1.72

sockeye

Upper Station Early

Gulf of Alaska

SST

-0.06

0.04

0.16
4.06
sockeye
Upper Station Late
Gulf of Alaska
SST
-0.10
0.01
0.11
0.63
sockeye
Karluk Early
Gulf of Alaska
SST
-0.14
-0.01
0.09
-0.74
sockeye
Karluk Late
Gulf of Alaska
SST
-0.09
0.02
0.12
1.68
sockeye
Ayakulik
Gulf of Alaska
SST
-0.06
0.04
0.15
4.47
sockeye
Chignik Early

Gulf of Alaska

SST

-0.19

-0.03

0.08

-2.71

sockeye

Chignik Lake

Gulf of Alaska

SST

-0.20

-0.03

0.08

-3.30

sockeye

Nelson River

Bering Sea

SST

-0.06

0.13

0.27

14.14

sockeye

Bear Lake

Bering Sea

SST

0.01

0.18

0.30

19.60

sockeye

Ugashik

Bering Sea

SST

0.17

0.30

0.50

35.29

sockeye

Egegik

Bering Sea

SST

0.07

0.21

0.34

23.35

sockeye

Igushik

Bering Sea

SST

0.07

0.20

0.32

22.17

sockeye

Naknek

Bering Sea

SST

0.10

0.21

0.32

23.48

sockeye

Nushagak

Bering Sea

SST

0.10

0.22

0.37

24.82

sockeye

Wood

Bering Sea

SST

0.15

0.24

0.35

27.18

sockeye

Alagnak

Bering Sea

SST

0.13

0.25

0.41

28.57

sockeye

Kvichak

Bering Sea

SST

-0.02

0.15

0.28

16.77

sockeye

Togiak

Bering Sea

SST

0.10

0.20

0.30

22.54

sockeye

Goodnews

Bering Sea

SST

0.05

0.20

0.35
22.63
sockeye
Birkenhead
West Coast
Comp
-0.49
-0.31
-0.18
-26.58
sockeye
Bowron
West Coast
Comp
-0.46
-0.30
-0.16
-25.55
sockeye
Chilko
West Coast
Comp
-0.39
-0.27
-0.12
-23.81
sockeye
Cultus
West Coast
Comp
-0.49
-0.31
-0.15
-26.69
sockeye
Early Stuart

West Coast

Comp

-0.43

-0.30

-0.19

-25.82

sockeye

Gates

West Coast

Comp

-0.42

-0.28

-0.11

-24.33

sockeye

Harrison

West Coast

Comp

-0.38

-0.25

-0.02

-21.77

sockeye

Late Shuswap

West Coast

Comp

-0.48

-0.30

-0.15

-26.23

sockeye

Late Stuart

West Coast

Comp

-0.46

-0.29

-0.15
-25.13
sockeye
Nadina
West Coast
Comp
-0.41
-0.27
-0.13
-23.97
sockeye
North Barriere
West Coast
Comp
-0.55
-0.33
-0.21
-28.21
sockeye
Pitt
West Coast
Comp
-0.41
-0.28
-0.16
-24.46
sockeye
Portage
West Coast
Comp
-0.53
-0.32
-0.18
-27.16
sockeye
Quesnel

West Coast

Comp

-0.50

-0.32

-0.20

-27.26

sockeye

Raft

West Coast

Comp

-0.44

-0.29

-0.17

-25.44

sockeye

Scotch

West Coast

Comp

-0.46

-0.30

-0.16

-25.96

sockeye

Seymour

West Coast

Comp

-0.46

-0.30

-0.16

-25.74

sockeye

Stellako

West Coast

Comp

-0.48

-0.31

-0.20
-27.01
sockeye
Weaver
West Coast
Comp
-0.46
-0.30
-0.15
-25.56
sockeye
Great Central Lake
West Coast
Comp
-0.36
-0.23
0.00
-20.82
sockeye
Sproat Lake
West Coast
Comp
-0.38
-0.26
-0.08
-22.82
sockeye
Skidegate
West Coast
Comp
-0.39
-0.27
-0.11
-23.32
sockeye
Evelyn

West Coast

Comp

-0.45

-0.29

-0.14

-25.51

sockeye

Awun

West Coast

Comp

-0.33

-0.18

0.09

-16.24

sockeye

Yakoun

West Coast

Comp

-0.41

-0.27

-0.15

-23.79

sockeye

Alastair

Southeast Alaska

Comp

-0.18

-0.04

0.10

-4.17

sockeye

Babine Early Wild

Southeast Alaska

Comp

-0.31

-0.10

0.03
-9.57
sockeye
Babine Late Wild
Southeast Alaska
Comp
-0.33
-0.11
0.03
-10.25
sockeye
Babine Mid Wild
Southeast Alaska
Comp
-0.30
-0.10
0.04
-9.12
sockeye
Bear-Skeena
Southeast Alaska
Comp
-0.12
0.00
0.17
-0.16
sockeye
Kitsumkalum
Southeast Alaska
Comp
-0.11
0.01
0.16
0.61
sockeye
Lakelse

Southeast Alaska

Comp

-0.17

-0.04

0.08

-4.15

sockeye

Morice

Southeast Alaska

Comp

-0.12

0.01

0.19

1.12

sockeye

Swan-Stephens

Southeast Alaska

Comp

-0.19

-0.05

0.08

-4.65

sockeye

Bowser

Southeast Alaska

Comp

-0.23

-0.08

0.04

-7.58

sockeye

Damdochax

Southeast Alaska

Comp

-0.22

-0.08

0.04
-7.27
sockeye
Kwinageese
Southeast Alaska
Comp
-0.21
-0.04
0.12
-4.23
sockeye
Lower Nass Sea & River Type
Southeast Alaska
Comp
-0.15
0.00
0.19
-0.47
sockeye
Meziadin
Southeast Alaska
Comp
-0.16
-0.05
0.07
-4.44
sockeye
Upper Nass River Type
Southeast Alaska
Comp
-0.19
-0.05
0.08
-5.19
sockeye
Redoubt

Southeast Alaska

Comp

-0.23

-0.06

0.09

-5.68

sockeye

Chilkoot

Southeast Alaska

Comp

-0.13

-0.01

0.16

-0.93

sockeye

Copper

Gulf of Alaska

Comp

-0.03

0.09

0.23

8.89

sockeye

Early Russian

Gulf of Alaska

Comp

-0.11

0.02

0.18

2.39

sockeye

Kenai

Gulf of Alaska

Comp

-0.24

-0.08

0.04
-7.36
sockeye
Kasilof
Gulf of Alaska
Comp
-0.06
0.03
0.15
3.55
sockeye
Frazer Lake
Gulf of Alaska
Comp
-0.20
-0.03
0.13
-3.17
sockeye
Upper Station Early
Gulf of Alaska
Comp
-0.10
0.02
0.16
2.24
sockeye
Upper Station Late
Gulf of Alaska
Comp
-0.17
-0.03
0.09
-3.06
sockeye
Karluk Early

Gulf of Alaska

Comp

-0.18

-0.04

0.08

-3.84

sockeye

Karluk Late

Gulf of Alaska

Comp

-0.13

-0.01

0.10

-1.06

sockeye

Ayakulik

Gulf of Alaska

Comp

-0.09

0.03

0.19

3.51

sockeye

Chignik Early

Gulf of Alaska

Comp

-0.22

-0.06

0.06

-5.85

sockeye

Chignik Late

Gulf of Alaska

Comp

-0.19

-0.05

0.06
-5.29
sockeye
Nelson River
Bering Sea
Comp
-0.16
0.00
0.11
-0.37
sockeye
Bear Lake
Bering Sea
Comp
-0.07
0.04
0.14
3.72
sockeye
Ugashik
Bering Sea
Comp
0.00
0.11
0.30
12.08
sockeye
Egegik
Bering Sea
Comp
-0.05
0.06
0.18
5.73
sockeye
Igushik

Bering Sea

Comp

-0.07

0.04

0.14

4.20

sockeye

Naknek

Bering Sea

Comp

-0.05

0.05

0.15

5.24

sockeye

Nushagak

Bering Sea

Comp

-0.07

0.05

0.16

4.98

sockeye

Wood

Bering Sea

Comp

-0.02

0.07

0.17

7.07

sockeye

Alagnak

Bering Sea

Comp

-0.03

0.08

0.21
7.99
sockeye
Kvichak
Bering Sea
Comp
-0.10
0.02
0.12
2.26
sockeye
Togiak
Bering Sea
Comp
-0.04
0.05
0.14
5.07
sockeye
Goodnews
Bering Sea
Comp
-0.08
0.04
0.16
4.59
sockeye
Birkenhead
West Coast
SST x Comp
-0.13
-0.02
0.10
-1.63
sockeye
Bowron

West Coast
 SST x Comp
 -0.12
 -0.01
 0.12
 -0.53
 sockeye
 Chilko
 West Coast
 SST x Comp
 -0.09
 0.01
 0.13
 0.59
 sockeye
 Cultus
 West Coast
 SST x Comp
 -0.14
 -0.01
 0.12
 -1.13
 sockeye
 Early Stuart
 West Coast
 SST x Comp
 -0.07
 0.03
 0.17
 2.94
 sockeye
 Gates
 West Coast
 SST x Comp
 -0.16
 -0.02

0.09
-2.45
sockeye
Harrison
West Coast
SST x Comp
-0.21
-0.04
0.07
-4.09
sockeye
Late Shuswap
West Coast
SST x Comp
-0.14
-0.02
0.10
-1.80
sockeye
Late Stuart
West Coast
SST x Comp
-0.11
0.00
0.14
-0.06
sockeye
Nadina
West Coast
SST x Comp
-0.10
0.01
0.15
0.64
sockeye
North Barriere

West Coast
 SST x Comp
 -0.10
 0.00
 0.13
 -0.04
 sockeye
 Pitt
 West Coast
 SST x Comp
 -0.18
 -0.05
 0.04
 -4.58
 sockeye
 Portage
 West Coast
 SST x Comp
 -0.12
 -0.01
 0.13
 -0.58
 sockeye
 Quesnel
 West Coast
 SST x Comp
 -0.14
 -0.02
 0.10
 -1.75
 sockeye
 Raft
 West Coast
 SST x Comp
 -0.15
 -0.03

0.07
-2.71
sockeye
Scotch
West Coast
SST x Comp
-0.12
0.00
0.14
-0.27
sockeye
Seymour
West Coast
SST x Comp
-0.13
-0.01
0.11
-1.26
sockeye
Stellako
West Coast
SST x Comp
-0.11
0.00
0.11
-0.42
sockeye
Weaver
West Coast
SST x Comp
-0.12
0.00
0.14
-0.20
sockeye
Great Central Lake

West Coast
 SST x Comp
 -0.18
 -0.04
 0.07
 -3.71
 sockeye
 Sproat Lake
 West Coast
 SST x Comp
 -0.22
 -0.05
 0.06
 -4.67
 sockeye
 Skidegate
 West Coast
 SST x Comp
 -0.12
 -0.01
 0.09
 -1.42
 sockeye
 Evelyn
 West Coast
 SST x Comp
 -0.12
 0.00
 0.14
 -0.13
 sockeye
 Awun
 West Coast
 SST x Comp
 -0.20
 -0.05

0.06
-4.47
sockeye
Yakoun
West Coast
SST x Comp
-0.13
-0.02
0.07
-2.27
sockeye
Alastair
Southeast Alaska
SST x Comp
-0.28
-0.09
0.01
-9.01
sockeye
Babine Early Wild
Southeast Alaska
SST x Comp
-0.15
-0.03
0.11
-2.91
sockeye
Babine Late Wild
Southeast Alaska
SST x Comp
-0.16
-0.04
0.09
-3.52
sockeye
Babine Mid Wild

Southeast Alaska

SST x Comp

-0.18

-0.05

0.06

-4.75

sockeye

Bear-Skeena

Southeast Alaska

SST x Comp

-0.13

-0.03

0.10

-2.74

sockeye

Kitsumkalum

Southeast Alaska

SST x Comp

-0.11

-0.02

0.08

-2.12

sockeye

Lakelse

Southeast Alaska

SST x Comp

-0.17

-0.05

0.05

-4.91

sockeye

Morice

Southeast Alaska

SST x Comp

-0.17

-0.05

0.06
 -4.95
 sockeye
 Swan-Stephens
 Southeast Alaska
 SST x Comp
 -0.16
 -0.05
 0.05
 -4.73
 sockeye
 Bowser
 Southeast Alaska
 SST x Comp
 -0.13
 -0.02
 0.10
 -2.47
 sockeye
 Damdochax
 Southeast Alaska
 SST x Comp
 -0.12
 -0.03
 0.08
 -2.59
 sockeye
 Kwinageese
 Southeast Alaska
 SST x Comp
 -0.16
 -0.04
 0.09
 -3.63
 sockeye
 Lower Nass Sea & River Type

Southeast Alaska

SST x Comp

-0.21

-0.06

0.06

-5.71

sockeye

Meziadin

Southeast Alaska

SST x Comp

-0.17

-0.06

0.03

-5.35

sockeye

Upper Nass River Type

Southeast Alaska

SST x Comp

-0.22

-0.07

0.03

-6.74

sockeye

Redoubt

Southeast Alaska

SST x Comp

-0.12

-0.01

0.15

-1.17

sockeye

Chilkoot

Southeast Alaska

SST x Comp

-0.11

-0.01

0.14
 -0.74
 sockeye
 Copper
 Gulf of Alaska
 SST x Comp
 -0.12
 -0.05
 0.01
 -4.48
 sockeye
 Early Russian
 Gulf of Alaska
 SST x Comp
 -0.10
 -0.03
 0.06
 -2.68
 sockeye
 Kenai
 Gulf of Alaska
 SST x Comp
 -0.09
 -0.03
 0.05
 -2.64
 sockeye
 Kasilof
 Gulf of Alaska
 SST x Comp
 -0.10
 -0.04
 0.01
 -3.93
 sockeye
 Frazer Lake

Gulf of Alaska
 SST x Comp
 -0.12
 -0.03
 0.05
 -3.27
 sockeye
 Upper Station Early
 Gulf of Alaska
 SST x Comp
 -0.10
 -0.03
 0.05
 -2.92
 sockeye
 Upper Station Late
 Gulf of Alaska
 SST x Comp
 -0.08
 -0.02
 0.07
 -1.90
 sockeye
 Karluk Early
 Gulf of Alaska
 SST x Comp
 -0.11
 -0.04
 0.03
 -3.72
 sockeye
 Karluk Late
 Gulf of Alaska
 SST x Comp
 -0.08
 -0.02

0.06
-2.07
sockeye
Ayakulik
Gulf of Alaska
SST x Comp
-0.09
-0.03
0.05
-2.58
sockeye
Chignik Early
Gulf of Alaska
SST x Comp
-0.14
-0.05
0.02
-4.80
sockeye
Chignik Late
Gulf of Alaska
SST x Comp
-0.11
-0.04
0.02
-3.78
sockeye
Nelson River
Bering Sea
SST x Comp
-0.11
-0.02
0.05
-1.80
sockeye
Bear Lake

Bering Sea
SST x Comp
-0.07
0.00
0.09
0.23
sockeye
Ugashik
Bering Sea
SST x Comp
-0.10
-0.02
0.05
-1.69
sockeye
Egegik
Bering Sea
SST x Comp
-0.09
-0.01
0.06
-1.05
sockeye
Igushik
Bering Sea
SST x Comp
-0.07
0.00
0.07
-0.08
sockeye
Naknek
Bering Sea
SST x Comp
-0.08
-0.01

0.05
-1.11
sockeye
Nushagak
Bering Sea
SST x Comp
-0.06
0.00
0.08
0.19
sockeye
Wood
Bering Sea
SST x Comp
-0.06
0.00
0.06
-0.04
sockeye
Alagnak
Bering Sea
SST x Comp
-0.07
0.00
0.07
-0.23
sockeye
Kvichak
Bering Sea
SST x Comp
-0.09
-0.01
0.05
-1.34
sockeye
Togiak

Bering Sea
 SST x Comp
 -0.05
 0.00
 0.07
 0.13
 sockeye
 Goodnews
 Bering Sea
 SST x Comp
 -0.09
 -0.01
 0.06
 -1.08
 chum
 Willapa Bay
 West Coast
 SST
 -0.23
 -0.10
 0.07
 -9.86
 chum
 Grays Harbour
 West Coast
 SST
 -0.24
 -0.10
 0.08
 -9.52
 chum
 S Sound Summer
 West Coast
 SST
 -0.23
 -0.10

0.09
-9.48
chum
S Sound Fall
West Coast
SST
-0.20
-0.08
0.09
-7.72
chum
S Sound Winter
West Coast
SST
-0.35
-0.17
-0.04
-15.79
chum
Hood Canal
West Coast
SST
-0.23
-0.09
0.08
-8.72
chum
Strait of Juan de Fuca
West Coast
SST
-0.53
-0.23
-0.07
-20.58
chum
Port Susan

West Coast

SST

-0.31

-0.14

0.00

-13.39

chum

Skagit

West Coast

SST

-0.31

-0.16

-0.02

-14.55

chum

Bellingham

West Coast

SST

-0.25

-0.12

0.00

-11.58

chum

Spiller-Fitz-Hugh-Burke

West Coast

SST

-0.22

-0.13

-0.04

-12.29

chum

Bella Coola-Dean Rivers

West Coast

SST

-0.17

-0.05

0.13
 -4.44
 chum
 Mussel-Kynoch
 West Coast
 SST
 -0.28
 -0.16
 -0.06
 -14.90
 chum
 Hecate Lowlands
 West Coast
 SST
 -0.26
 -0.16
 -0.06
 -14.62
 chum
 Douglas-Gardner
 West Coast
 SST
 -0.34
 -0.18
 -0.06
 -16.24
 chum
 Afognak
 Gulf of Alaska
 SST
 -0.24
 0.03
 0.28
 3.35
 chum
 E Kodiak

Gulf of Alaska

SST

-0.12

0.03

0.19

3.50

chum

NW Kodiak

Gulf of Alaska

SST

-0.13

0.00

0.11

0.30

chum

Alitak

Gulf of Alaska

SST

-0.12

0.03

0.18

3.39

chum

SW Kodiak

Gulf of Alaska

SST

-0.25

0.00

0.19

-0.39

chum

Mainland Kodiak

Gulf of Alaska

SST

-0.09

0.04

0.19
4.01
chum
Chignik
Gulf of Alaska
SST
-0.07
0.06
0.23
5.69
chum
NW AK Peninsula
Bering Sea
SST
-0.06
0.03
0.13
3.35
chum
S AK Peninsula
Bering Sea
SST
0.00
0.09
0.20
9.89
chum
N AK Peninsula
Bering Sea
SST
-0.06
0.04
0.14
3.84
chum
Yukon - Summer

Bering Sea
SST
-0.21
-0.03
0.10
-2.80
chum
Yukon - Fall
Bering Sea
SST
-0.18
-0.01
0.11
-0.64
chum
Kwiniuk Inlet
Bering Sea
SST
-0.17
-0.01
0.11
-1.45
chum
Willapa Bay
West Coast
Comp
-0.21
-0.09
0.01
-8.69
chum
Grays Harbour
West Coast
Comp
-0.20
-0.08

0.03
-8.01
chum
S Sound Summer
West Coast
Comp
-0.22
-0.09
0.01
-8.74
chum
S Sound Fall
West Coast
Comp
-0.16
-0.07
0.06
-6.48
chum
S Sound Winter
West Coast
Comp
-0.24
-0.12
-0.02
-10.92
chum
Hood Canal
West Coast
Comp
-0.15
-0.06
0.11
-5.59
chum
Strait of Juan de Fuca

West Coast

Comp

-0.25

-0.12

0.01

-11.41

chum

Port Susan

West Coast

Comp

-0.22

-0.10

0.00

-9.73

chum

Skagit

West Coast

Comp

-0.27

-0.12

-0.02

-11.13

chum

Bellingham

West Coast

Comp

-0.17

-0.08

0.04

-7.79

chum

Spiller-Fitz-Hugh-Burke

West Coast

Comp

-0.18

-0.10

-0.02
 -9.19
 chum
 Bella Coola-Dean Rivers
 West Coast
 Comp
 -0.14
 -0.05
 0.10
 -4.43
 chum
 Mussel-Kynoch
 West Coast
 Comp
 -0.19
 -0.10
 -0.01
 -9.66
 chum
 Hecate Lowlands
 West Coast
 Comp
 -0.25
 -0.12
 -0.04
 -11.51
 chum
 Douglas-Gardner
 West Coast
 Comp
 -0.21
 -0.11
 0.01
 -9.97
 chum
 Afognak

Gulf of Alaska

Comp

-0.11

0.07

0.38

7.42

chum

E Kodiak

Gulf of Alaska

Comp

-0.06

0.06

0.21

5.95

chum

NW Kodiak

Gulf of Alaska

Comp

-0.08

0.02

0.12

2.35

chum

Alitak

Gulf of Alaska

Comp

-0.05

0.07

0.23

6.84

chum

SW Kodiak

Gulf of Alaska

Comp

-0.24

0.00

0.16
0.46
chum
Mainland Kodiak
Gulf of Alaska
Comp
-0.09
0.04
0.16
4.09
chum
Chignik
Gulf of Alaska
Comp
-0.10
0.04
0.16
4.37
chum
NW AK Peninsula
Bering Sea
Comp
-0.15
-0.02
0.11
-1.81
chum
S AK Peninsula
Bering Sea
Comp
-0.04
0.10
0.25
10.28
chum
N AK Peninsula

Bering Sea
 Comp
 -0.16
 -0.02
 0.12
 -1.76
 chum
 Yukon - Summer
 Bering Sea
 Comp
 -0.34
 -0.13
 0.05
 -11.90
 chum
 Yukon - Fall
 Bering Sea
 Comp
 -0.24
 -0.07
 0.10
 -6.86
 chum
 Kwiniuk Inlet
 Bering Sea
 Comp
 -0.38
 -0.13
 0.07
 -11.83
 chum
 Willapa Bay
 West Coast
 SST x Comp
 -0.20
 -0.10

-0.01
-9.37
chum
Grays Harbour
West Coast
SST x Comp
-0.19
-0.10
0.00
-9.10
chum
S Sound Summer
West Coast
SST x Comp
-0.19
-0.09
0.00
-8.93
chum
S Sound Fall
West Coast
SST x Comp
-0.19
-0.10
-0.01
-9.23
chum
S Sound Winter
West Coast
SST x Comp
-0.21
-0.10
-0.02
-9.64
chum
Hood Canal

West Coast
 SST x Comp
 -0.22
 -0.10
 -0.02
 -9.94
 chum
 Strait of Juan de Fuca
 West Coast
 SST x Comp
 -0.23
 -0.11
 -0.02
 -10.06
 chum
 Port Susan
 West Coast
 SST x Comp
 -0.21
 -0.10
 -0.02
 -9.82
 chum
 Skagit
 West Coast
 SST x Comp
 -0.17
 -0.08
 0.03
 -7.88
 chum
 Bellingham
 West Coast
 SST x Comp
 -0.25
 -0.12

-0.04
 -11.29
 chum
 Spiller-Fitz-Hugh-Burke
 West Coast
 SST x Comp
 -0.15
 -0.08
 -0.01
 -8.00
 chum
 Bella Coola-Dean Rivers
 West Coast
 SST x Comp
 -0.16
 -0.08
 0.02
 -7.78
 chum
 Mussel-Kynoch
 West Coast
 SST x Comp
 -0.17
 -0.09
 -0.02
 -8.74
 chum
 Hecate Lowlands
 West Coast
 SST x Comp
 -0.16
 -0.09
 -0.01
 -8.36
 chum
 Douglas-Gardner

West Coast
 SST x Comp
 -0.17
 -0.09
 0.00
 -8.34
 chum
 Afognak
 Gulf of Alaska
 SST x Comp
 -0.11
 0.07
 0.26
 6.79
 chum
 E Kodiak
 Gulf of Alaska
 SST x Comp
 -0.06
 0.06
 0.18
 6.38
 chum
 NW Kodiak
 Gulf of Alaska
 SST x Comp
 -0.02
 0.07
 0.16
 6.92
 chum
 Alitak
 Gulf of Alaska
 SST x Comp
 -0.04
 0.07

0.20
7.55
chum
SW Kodiak
Gulf of Alaska
SST x Comp
-0.04
0.10
0.33
10.33
chum
Mainland Kodiak
Gulf of Alaska
SST x Comp
-0.05
0.06
0.16
6.07
chum
Chignik
Gulf of Alaska
SST x Comp
-0.10
0.03
0.13
3.41
chum
NW AK Peninsula
Bering Sea
SST x Comp
-0.13
-0.04
0.05
-4.37
chum
S AK Peninsula

Bering Sea
 SST x Comp
 -0.13
 -0.05
 0.04
 -4.60
 chum
 N AK Peninsula
 Bering Sea
 SST x Comp
 -0.18
 -0.07
 0.02
 -6.93
 chum
 Yukon - Summer
 Bering Sea
 SST x Comp
 -0.26
 -0.09
 0.02
 -8.18
 chum
 Yukon - Fall
 Bering Sea
 SST x Comp
 -0.17
 -0.06
 0.04
 -6.08
 chum
 Kwiniuk Inlet
 Bering Sea
 SST x Comp
 -0.19
 -0.06

0.05
-6.10
pink
Nisqually-Odd
West Coast
SST
-0.28
-0.13
0.02
-12.49
pink
S South Misc.-Odd
West Coast
SST
-0.30
-0.15
-0.01
-13.73
pink
Puyallup-Odd
West Coast
SST
-0.27
-0.13
0.01
-12.24
pink
Hood Canal-Odd
West Coast
SST
-0.27
-0.13
0.02
-12.34
pink
Snohomish-Odd

West Coast
SST
-0.25
-0.11
0.04
-10.73
pink
Dungeness-Odd
West Coast
SST
-0.27
-0.13
0.02
-12.21
pink
Stillaguamish-Odd
West Coast
SST
-0.26
-0.12
0.02
-11.74
pink
Skagit-Odd
West Coast
SST
-0.26
-0.13
0.00
-12.31
pink
Nooksack-Odd
West Coast
SST
-0.27
-0.13

0.01
-12.37
pink
Fraser-Odd
West Coast
SST
-0.26
-0.12
0.05
-11.54
pink
Hecate Lowlands-Even
West Coast
SST
-0.32
-0.18
-0.05
-16.29
pink
Hecate Strait-Fjords-Even
West Coast
SST
-0.30
-0.16
-0.03
-15.17
pink
N Haida Gwaii-Even
Southeast Alaska
SST
-0.26
-0.02
0.20
-2.07
pink
S Southeast Alaska-Even

Southeast Alaska

SST

-0.19

-0.02

0.13

-2.45

pink

S Southeast Alaska-Odd

Southeast Alaska

SST

-0.09

0.08

0.26

8.37

pink

N Southeast Alaska Outside-Even

Southeast Alaska

SST

-0.13

0.05

0.24

5.21

pink

N Southeast Alaska Outside-Odd

Southeast Alaska

SST

-0.08

0.11

0.34

11.70

pink

N Southeast Alaska Inside-Even

Southeast Alaska

SST

-0.18

0.01

0.21
0.51
pink
N Southeast Alaska Inside-Odd
Southeast Alaska
SST
-0.06
0.12
0.35
13.10
pink
Auke Creek-Even
Southeast Alaska
SST
-0.69
-0.26
0.08
-22.77
pink
Auke Creek-Odd
Southeast Alaska
SST
-0.67
-0.25
0.09
-21.74
pink
Prince William Sound-Even
Gulf of Alaska
SST
0.09
0.27
0.46
31.63
pink
Prince William Sound-Odd

Gulf of Alaska

SST

-0.02

0.17

0.35

18.68

pink

Kodiak-Even

Gulf of Alaska

SST

0.05

0.23

0.41

26.18

pink

Kodiak-Odd

Gulf of Alaska

SST

0.06

0.25

0.44

28.61

pink

Chignik-Even

Gulf of Alaska

SST

-0.23

0.02

0.28

2.33

pink

Chignik-Odd

Gulf of Alaska

SST

0.21

0.38

0.54
45.67
pink
S AKPen-Even
Gulf of Alaska
SST
0.07
0.30
0.56
35.18
pink
S AKPen-Odd
Gulf of Alaska
SST
0.33
0.59
0.85
81.07
pink
NW AK Peninsula-Even
Bering Sea
SST
0.10
0.49
0.93
64.00
pink
NW AK Peninsula-Odd
Bering Sea
SST
0.18
0.55
0.95
73.20
pink
Kwiniuk-Even

Bering Sea
SST
0.17
0.49
0.81
63.82
pink
Kwiniuk-Odd
Bering Sea
SST
0.12
0.53
0.98
69.17
pink
Nisqually-Odd
West Coast
Comp
-0.39
0.00
0.44
-0.36
pink
S South Misc.-Odd
West Coast
Comp
-0.42
-0.09
0.25
-8.90
pink
Puyallup-Odd
West Coast
Comp
-0.31
0.01

0.38
0.68
pink
Hood Canal-Odd
West Coast
Comp
-0.38
-0.04
0.32
-3.90
pink
Snohomish-Odd
West Coast
Comp
-0.20
0.13
0.50
13.73
pink
Dungeness-Odd
West Coast
Comp
-0.36
0.00
0.39
-0.39
pink
Stillaguamish-Odd
West Coast
Comp
-0.25
0.04
0.36
3.70
pink
Skagit-Odd

West Coast

Comp

-0.28

-0.01

0.27

-1.37

pink

Nooksack-Odd

West Coast

Comp

-0.33

-0.05

0.25

-4.65

pink

Fraser-Odd

West Coast

Comp

-0.32

-0.03

0.27

-2.89

pink

Hecate Lowlands-Even

West Coast

Comp

-0.60

-0.36

-0.09

-30.08

pink

Hecate Strait-Fjords-Even

West Coast

Comp

-0.62

-0.30

0.00
 -25.81
 pink
 N Haida Gwaii-Even
 Southeast Alaska
 Comp
 -0.04
 0.16
 0.32
 17.00
 pink
 S Southeast Alaska-Even
 Southeast Alaska
 Comp
 0.01
 0.16
 0.29
 16.94
 pink
 S Southeast Alaska-Odd
 Southeast Alaska
 Comp
 0.09
 0.23
 0.42
 26.06
 pink
 N Southeast Alaska Outside-Even
 Southeast Alaska
 Comp
 0.05
 0.21
 0.39
 22.93
 pink
 N Southeast Alaska Outside-Odd

Southeast Alaska
 Comp
 0.08
 0.23
 0.41
 25.89
 pink
 N Southeast Alaska Inside-Even
 Southeast Alaska
 Comp
 -0.02
 0.16
 0.31
 17.46
 pink
 N Southeast Alaska Inside-Odd
 Southeast Alaska
 Comp
 0.09
 0.24
 0.42
 27.20
 pink
 Auke Creek-Even
 Southeast Alaska
 Comp
 -0.26
 0.04
 0.26
 4.41
 pink
 Auke Creek-Odd
 Southeast Alaska
 Comp
 -0.24
 0.05

0.26
5.18
pink
Prince William Sound-Even
Gulf of Alaska
Comp
-0.13
-0.01
0.12
-0.69
pink
Prince William Sound-Odd
Gulf of Alaska
Comp
-0.21
-0.06
0.07
-5.54
pink
Kodiak-Even
Gulf of Alaska
Comp
-0.18
-0.03
0.10
-3.06
pink
Kodiak-Odd
Gulf of Alaska
Comp
-0.14
-0.01
0.13
-0.91
pink
Chignik-Even

Gulf of Alaska

Comp

-0.38

-0.14

0.04

-13.30

pink

Chignik-Odd

Gulf of Alaska

Comp

-0.08

0.04

0.17

3.81

pink

S AKPen-Even

Gulf of Alaska

Comp

-0.17

0.00

0.16

-0.45

pink

S AKPen-Odd

Gulf of Alaska

Comp

-0.04

0.16

0.42

17.43

pink

NW AK Peninsula-Even

Bering Sea

Comp

-0.48

0.06

0.50
6.25
pink
NW AK Peninsula-Odd
Bering Sea
Comp
-0.20
0.22
0.73
24.12
pink
Kwiniuk-Even
Bering Sea
Comp
-0.23
0.14
0.54
15.28
pink
Kwiniuk-Odd
Bering Sea
Comp
-0.39
0.10
0.55
10.70
pink
Nisqually-Odd
West Coast
SST x Comp
-0.50
-0.20
0.05
-18.47
pink
S South Misc.-Odd

West Coast
 SST x Comp
 -0.58
 -0.25
 -0.02
 -22.28
 pink
 Puyallup-Odd
 West Coast
 SST x Comp
 -0.40
 -0.17
 0.07
 -15.40
 pink
 Hood Canal-Odd
 West Coast
 SST x Comp
 -0.35
 -0.12
 0.20
 -10.97
 pink
 Snohomish-Odd
 West Coast
 SST x Comp
 -0.45
 -0.21
 0.00
 -18.74
 pink
 Dungeness-Odd
 West Coast
 SST x Comp
 -0.49
 -0.20

0.05
 -18.34
 pink
 Stillaguamish-Odd
 West Coast
 SST x Comp
 -0.52
 -0.24
 -0.03
 -21.46
 pink
 Skagit-Odd
 West Coast
 SST x Comp
 -0.51
 -0.24
 -0.04
 -21.40
 pink
 Nooksack-Odd
 West Coast
 SST x Comp
 -0.46
 -0.21
 0.00
 -19.25
 pink
 Fraser-Odd
 West Coast
 SST x Comp
 -0.29
 -0.07
 0.20
 -7.19
 pink
 Hecate Lowlands-Even

West Coast
 SST x Comp
 -0.31
 -0.14
 0.04
 -13.39
 pink
 Hecate Strait-Fjords-Even
 West Coast
 SST x Comp
 -0.38
 -0.16
 0.07
 -15.03
 pink
 N Haida Gwaii-Even
 Southeast Alaska
 SST x Comp
 -0.29
 -0.14
 0.01
 -13.09
 pink
 S Southeast Alaska-Even
 Southeast Alaska
 SST x Comp
 -0.26
 -0.14
 -0.01
 -13.11
 pink
 S Southeast Alaska-Odd
 Southeast Alaska
 SST x Comp
 -0.29
 -0.15

-0.03
 -14.27
 pink
 N Southeast Alaska Outside-Even
 Southeast Alaska
 SST x Comp
 -0.24
 -0.12
 0.04
 -11.16
 pink
 N Southeast Alaska Outside-Odd
 Southeast Alaska
 SST x Comp
 -0.28
 -0.14
 0.00
 -13.05
 pink
 N Southeast Alaska Inside-Even
 Southeast Alaska
 SST x Comp
 -0.30
 -0.15
 -0.01
 -13.83
 pink
 N Southeast Alaska Inside-Odd
 Southeast Alaska
 SST x Comp
 -0.30
 -0.15
 -0.02
 -14.16
 pink
 Auke Creek-Even

Southeast Alaska

SST x Comp

-0.30

-0.13

0.06

-12.41

pink

Auke Creek-Odd

Southeast Alaska

SST x Comp

-0.36

-0.16

-0.01

-14.80

pink

Prince William Sound-Even

Gulf of Alaska

SST x Comp

-0.36

-0.15

0.05

-14.27

pink

Prince William Sound-Odd

Gulf of Alaska

SST x Comp

-0.11

0.09

0.28

9.20

pink

Kodiak-Even

Gulf of Alaska

SST x Comp

-0.21

0.01

0.26
 0.95
 pink
 Kodiak-Odd
 Gulf of Alaska
 SST x Comp
 -0.29
 -0.10
 0.09
 -9.92
 pink
 Chignik-Even
 Gulf of Alaska
 SST x Comp
 -0.26
 -0.01
 0.27
 -0.74
 pink
 Chignik-Odd
 Gulf of Alaska
 SST x Comp
 -0.38
 -0.21
 -0.04
 -18.90
 pink
 S AKPen-Even
 Gulf of Alaska
 SST x Comp
 -0.48
 -0.23
 0.01
 -20.32
 pink
 S AKPen-Odd

Gulf of Alaska
 SST x Comp
 -0.46
 -0.24
 -0.03
 -21.53
 pink
 NW AK Peninsula-Even
 Bering Sea
 SST x Comp
 -0.58
 -0.17
 0.31
 -15.39
 pink
 NW AK Peninsula-Odd
 Bering Sea
 SST x Comp
 -0.72
 -0.27
 0.13
 -23.82
 pink
 Kwiniuk-Even
 Bering Sea
 SST x Comp
 -0.54
 -0.27
 0.00
 -23.52
 pink
 Kwiniuk-Odd
 Bering Sea
 SST x Comp
 -0.52
 -0.03

0.71

-2.70

Table S4. Estimated regional mean effect sizes for SST and Competitor covariates, from time-varying ‘era’ models.

Species

Ocean Region

Coefficient

Era

Lower CI

Mean

Upper CI

sockeye

West Coast

SST

Early

-0.10

0.00

0.10

sockeye

Southeast Alaska

SST

Early

-0.08

0.06

0.20

sockeye

Gulf of Alaska

SST

Early

-0.01

0.10

0.20

sockeye

Bering Sea

SST

Early

0.08
0.19
0.30
sockeye
West Coast
SST
Middle
-0.22
-0.13
-0.04
sockeye
Southeast Alaska
SST
Middle
-0.05
0.05
0.16
sockeye
Gulf of Alaska
SST
Middle
-0.06
0.05
0.15
sockeye
Bering Sea
SST
Middle
0.16
0.27
0.39
sockeye
West Coast
SST
Late
-0.47

-0.31
-0.15
sockeye
Southeast Alaska
SST
Late
-0.44
-0.29
-0.14
sockeye
Gulf of Alaska
SST
Late
-0.28
-0.14
-0.01
sockeye
Bering Sea
SST
Late
0.05
0.18
0.30
sockeye
West Coast
Competitors
Stationary
-0.35
-0.28
-0.20
sockeye
Southeast Alaska
Competitors
Stationary
-0.11
-0.03

0.05
sockeye
Gulf of Alaska
Competitors
Stationary
-0.09
0.00
0.08
sockeye
Bering Sea
Competitors
Stationary
-0.02
0.06
0.14
chum
West Coast
SST
Early
-0.15
-0.05
0.04
chum
Gulf of Alaska
SST
Early
-0.42
-0.14
0.11
chum
Bering Sea
SST
Early
-0.02
0.12
0.24

chum
West Coast
SST
Middle
-0.27
-0.18
-0.10
chum
Gulf of Alaska
SST
Middle
-0.20
0.00
0.17
chum
Bering Sea
SST
Middle
-0.09
0.06
0.21
chum
West Coast
SST
Late
-0.46
-0.32
-0.17
chum
Gulf of Alaska
SST
Late
-0.09
0.09
0.29
chum

Bering Sea
SST
Late
-0.23
-0.09
0.04
chum
West Coast
Competitors
Stationary
-0.16
-0.08
-0.01
chum
Gulf of Alaska
Competitors
Stationary
-0.04
0.07
0.19
chum
Bering Sea
Competitors
Stationary
-0.26
-0.06
0.10
pink
West Coast
SST
Early
-0.27
-0.03
0.20
pink
Southeast Alaska

SST
Early
0.03
0.18
0.33
pink
Gulf of Alaska
SST
Early
0.18
0.41
0.63
pink
Bering Sea
SST
Early
0.45
1.09
1.70
pink
West Coast
SST
Middle
-0.49
-0.29
-0.11
pink
Southeast Alaska
SST
Middle
-0.15
0.02
0.19
pink
Gulf of Alaska
SST

Middle
-0.05
0.20
0.45
pink
Bering Sea
SST
Middle
-0.20
0.38
0.99
pink
West Coast
SST
Late
-0.64
-0.38
-0.11
pink
Southeast Alaska
SST
Late
-0.38
-0.21
-0.05
pink
Gulf of Alaska
SST
Late
-0.08
0.17
0.40
pink
Bering Sea
SST
Late

-0.25
 0.27
 0.84
 pink
 West Coast
 Competitors
 Stationary
 -0.22
 0.00
 0.22
 pink
 Southeast Alaska
 Competitors
 Stationary
 0.02
 0.17
 0.32
 pink
 Gulf of Alaska
 Competitors
 Stationary
 -0.19
 -0.03
 0.13
 pink
 Bering Sea
 Competitors
 Stationary
 -0.57
 0.00
 0.56

Table S5. Estimated stock-specific mean effect sizes for SST and Competitor covariates, from time-varying ‘era’ models.

- need to figure out what to present, not worth doing all years is it?

Table S6. Estimated regional mean effect sizes for SST and Competitor covariates, from annually time-varying (random walk) models. - need to figure out what to present, not worth doing all years is it?

Table S8. Priors on all directly estimated parameters for each model type.

See: *Output/Model_priors.xlsx*

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